



Technical Service Bulletin

Technical Service Bulletin: tsb100938

Released Date: 07-Jan-2015

Oversize Cylinder Liners

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Service/Parts Topic obsoletes and supersedes Topic 10T1-32, Topic 05T1-25, and Topic 06T1-19.

This Service/Parts Topic informs the field of oversize cylinder liners for use on the QSK45 and QSK60 Series engines.

The cylinder liners have been released for use when there is a requirement to machine the upper/lower press fit and or counterbore ledge due to any wear or fretting damage that may have occurred. During engine rebuild, the upper/lower press fit areas and counterbore ledge of the cylinder block should be inspected for any evidence of wear or fretting. Any cylinder bores that show signs of wear and/or fretting should be machined and an appropriate oversize cylinder liner fitted. Engine block upper and lower press fits that show no signs of wear or fretting need **not** have oversize cylinder liners fitted and standard cylinder liners can be used.

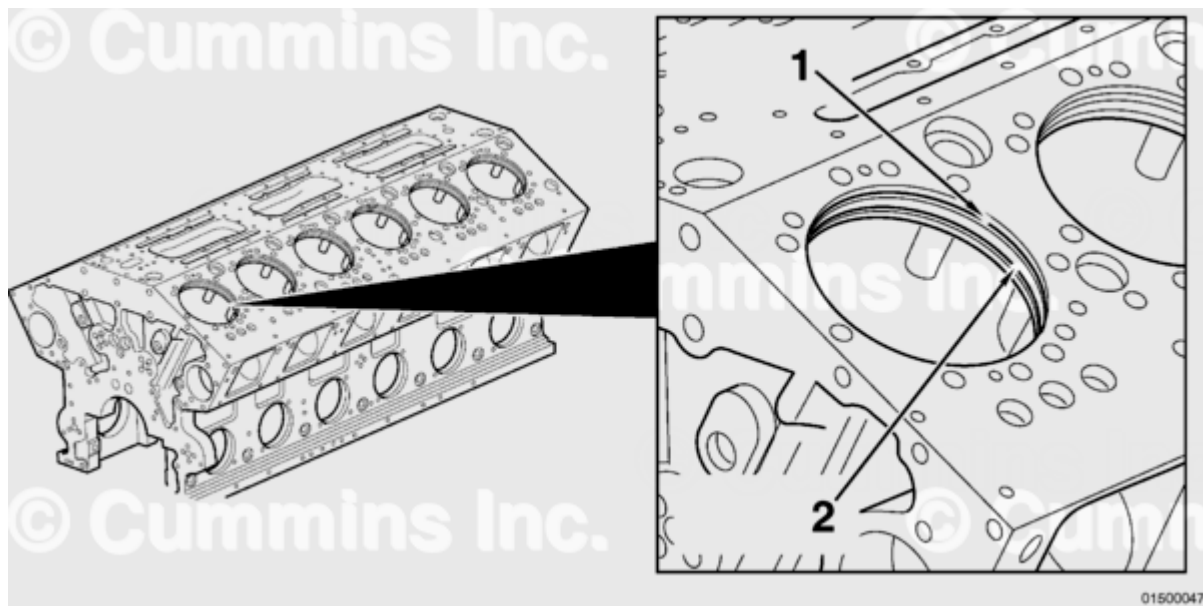


Figure 1

See Figure 1 for upper press fit diameter (2) location and the upper press fit seating ledge (1) location.

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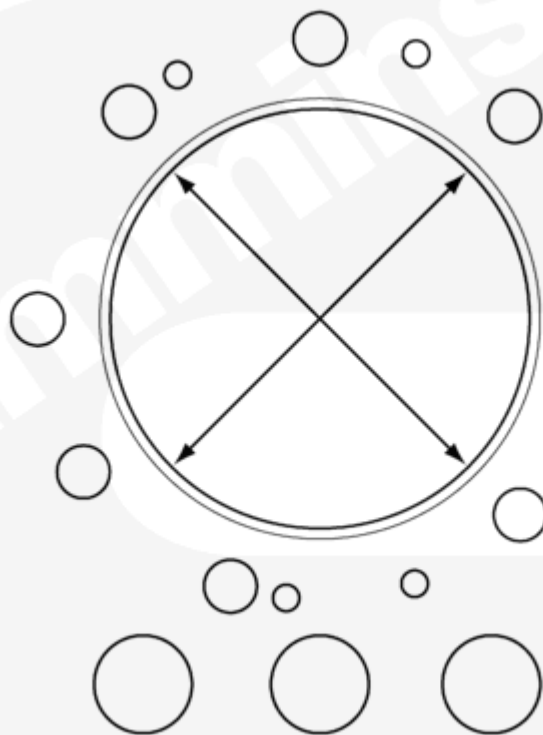


Figure 2

Cylinder block upper and lower press fit bores should be checked for wear across the diameter at the locations shown above. See Figure 2. See Table 1 for wear specifications.

Table 1: Cylinder Upper and Lower Bore Specifications

Upper Press Fit			Lower Press Fit		
mm		in	mm		in

Table 1: Cylinder Upper and Lower Bore Specifications						
Upper Press Fit				Lower Press Fit		
190.28	MIN	7.491		181.74	MIN	7.155
190.34	MAX	7.494		181.80	MAX	7.157

Three oversize cylinder liner parts along with oversize cylinder liner shims for the respective oversize cylinder liners have been released.

- 0.25 mm [0.010 in]
- 0.51 mm [0.020 in]
- 0.76 mm [0.030 in]

Standard non oversize cylinder liner shims can be used with both the standard cylinder liner and for the 0.25 mm [0.010 in] oversize cylinder liner. A 0.51 mm [0.020 in] oversize cylinder liner shim can be used for both the 0.51 mm [0.020 in] and 0.76 mm [0.030 in] oversize cylinder liners The oversize dimensional features of the cylinder liner apply to the upper press fit diameter, lower press fit diameter, and the upper press fit seating ledge. All other dimensional features of the cylinder liner remain unchanged. For available cylinder liner shims see Table 3.

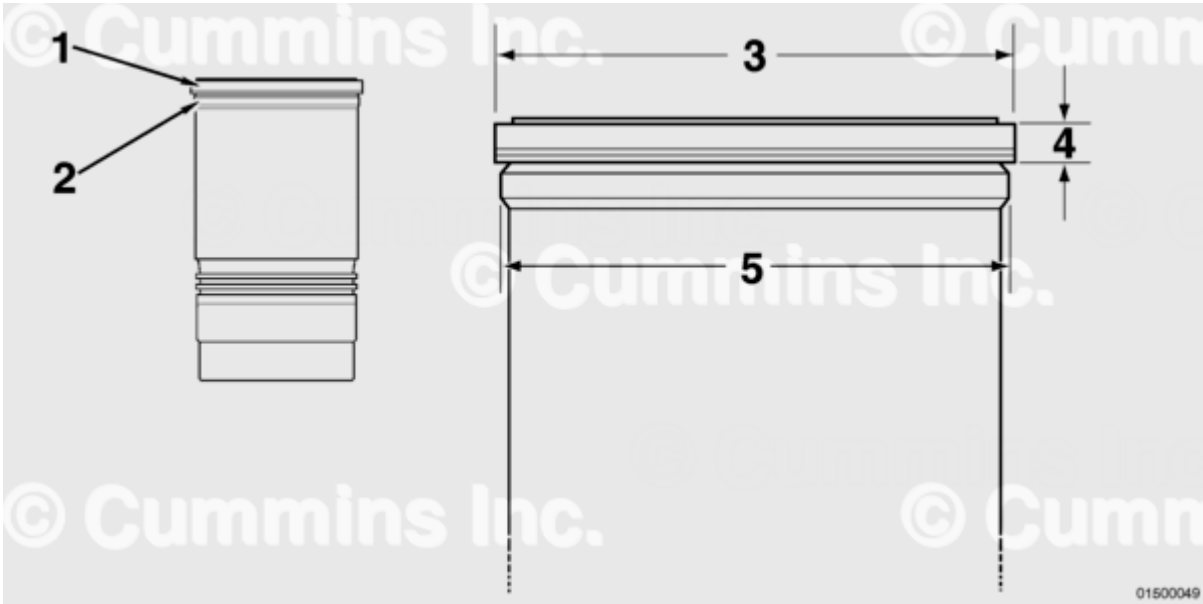


Figure 3

The upper press fit (1 and 3), lower press fit (2 and 5) and the seating face dimensions (4) all increase by the respective cylinder liner oversize value. The 0.25 mm [0.010 in] oversize cylinder liner, has the upper press fit. The lower press fit and the upper press fit seating ledge dimensions all increased by 0.25 mm [0.010 in] from their nominal dimension values. See Figure 3. See Table 2 and 3.

Table 2: Cylinder Liner Key Dimensions									
Cylind er Liner Part Numb er	Cylind er Liner Kit Part Numb er	Oversi ze Value	Upper Press Fit Diameter (3)		Lower Press Fit Diameter (5)		Upper Press Fit Seating Ledge Depth (4)		Cylind er Liner Shim Requir ement
			MIN	MAX	MIN	MAX	MIN	MAX	
40859 17	40891 43	Stand ard Cylind er Liner	190.3 1 mm [7.493 in]	190.3 6 mm [7.495 in]	181.8 2 mm [7.158 in]	181.8 6 mm [7.160 in]	13.36 mm [0.526 in]	13.39 mm [0.527 in]	Stand ard
36369 98	40899 91	0.25 mm [0.010 in]	190.5 6 mm [7.502 in]	190.6 1 mm [7.504 in]	182.0 7 mm [7.168 in]	182.1 1 mm [7.170 in]	13.61 mm [0.536 in]	13.64 mm [0.537 in]	Stand ard
36369 99	40899 92	0.51 mm [0.020 in]	190.8 1 mm [7.512 in]	190.8 6 mm [7.514 in]	182.3 2 mm [7.178 in]	182.3 6 mm [7.180 in]	13.86 mm [0.546 in]	13.89 mm [0.547 in]	0.51 mm [0.020 in]
36370 00	40899 93	0.76 mm [0.030 in]	191.0 6 mm [7.522 in]	191.1 1 mm [7.524 in]	182.5 7 mm [7.188 in]	182.6 1 mm [7.190 in]	14.11 mm [0.556 in]	14.14 mm [0.557 in]	0.51 mm [0.020 in]

Table 3: Cylinder Block Key Dimensions									
Cylind er Liner Part Numb er	Cylind er Liner Kit Part Numb er	Oversi ze Value	Upper Press Fit Diameter (1)		Lower Press Fit Diameter (2)		Upper Press Fit Seating Ledge Depth in Block		Cylind er Liner Shim Requir ement
			MIN	MAX	MIN	MAX	MIN	MAX	
40859 17	40891 43	Stand ard Cylind er Liner	190.2 8 mm [7.491 in]	190.3 4 mm [7.495 in]	181.7 4 mm [7.155 in]	181.8 0 mm [7.157 in]	13.68 mm [0.538 in]	13.74 mm [0.541 in]	Stand ard

Table 3: Cylinder Block Key Dimensions									
Cylinder Liner Part Number	Cylinder Liner Kit Part Number	Oversize Value	Upper Press Fit Diameter (1)		Lower Press Fit Diameter (2)		Upper Press Fit Seating Ledge Depth in Block		Cylinder Liner Shim Requirement
			MIN	MAX	MIN	MAX	MIN	MAX	
3636998	4089991	0.25 mm [0.010 in]	190.53 mm [7.501 in]	190.59 mm [7.504 in]	181.99 mm [7.165 in]	182.05 mm [7.167 in]	13.93 mm [0.548 in]	13.99 mm [0.551 in]	Standard
3636999	4089992	0.51 mm [0.020 in]	190.78 mm [7.511 in]	190.84 mm [7.514 in]	182.24 mm [7.175 in]	182.30 mm [7.177 in]	14.18 mm [0.558 in]	14.24 mm [0.561 in]	0.51 mm [0.020 in]
3637000	4089993	0.76 mm [0.030 in]	191.04 mm [7.521 in]	191.10 mm [7.524 in]	182.50 mm [7.185 in]	182.56 mm [7.187 in]	14.44 mm [0.568 in]	14.5 mm [0.571 in]	0.51 mm [0.020 in]

Various shim thicknesses are available for both the standard and the 0.51 mm [0.020 in] oversize shim to enable the correct cylinder liner protrusion to be obtained upon engine assembly.

If a standard (non oversize) cylinder liner is to be fitted to an engine block, no work needs to be carried out on the engine block. If an oversize cylinder liner is to be fitted, the engine block **must** be machined for the respective oversize cylinder liner. The upper press fit (1), lower press fit (2), and upper press fit seating face features **must** all be machined to enable the correct fit of an oversize cylinder liner. See Figure 1. Any number of cylinder bores can be machined for oversize cylinder liners.

Table 4: Shim Part Numbers		
Cylinder Liner Part Number	Shim Part Number	Shim Thickness
4085917 and 3636998	3088385	0.45 mm [0.0180 in]
	3088386	0.52 mm [0.0205 in]
	3088387	0.55 mm [0.0220 in]
	3331919	0.60 mm [0.0240 in]
	3088388	0.78 mm [0.0310 in]

Table 4: Shim Part Numbers		
Cylinder Liner Part Number	Shim Part Number	Shim Thickness
3636999 and 3637000	3637009	0.45 mm [0.0180 in]
	3637008	0.52 mm [0.0205 in]
	3637010	0.55 mm [0.0220 in]
	3637011	0.60 mm [0.0240 in]
	3637012	0.78 mm [0.0310 in]

Document History

Date	Details
2014-11-26	none
2015-1-7	none